

A Formula Atlas for the Thue–Morse Sequence

Digital, valiative, Pascal, generating, Prouhet, Fourier, spectral, and arithmetic representations

Consolidated companion report to the Fabius–Rvachev formal development

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26 August 2026

Abstract

Four independent drafts in the ProveIt Fabius-function research-frontier directory studied exact formulae for the Thue–Morse sequence from overlapping but nonidentical viewpoints. This article merges them into one coherent exposition, removes duplicated proofs, reconciles notation and normalizations, and retains the strongest version of each result. It develops the signed sequence $\varepsilon_n = (-1)^{s_2(n)}$ and the zero-one bit $\tau(n) = s_2(n) \bmod 2$ through binary recurrences, floor and fractional-part sums, trigonometric signs, factorial and binomial valuations, carry cocycles, Pascal support, multinomial parity, Boolean Möbius inversion, odious and evil enumerators, lacunary Euler products, nonlinear recurrences, Bell polynomials, Hessenberg determinants, algebraic power series over $\mathbb{F}_2$, finite differences, exact Prouhet moments, sparse Pascal-row cancellation, ordinary and Walsh Fourier transforms, Riesz products, autocorrelation, Dirichlet transforms, and evaluated infinite products.

Several additions go beyond all four drafts. We derive exact dyadic formulas for Thue–Morse sums in residue classes modulo three and place them inside Newman’s phenomenon; record the overlap-free and substitution-dynamical structure of the infinite word; explain the nonvanishing Hankel-determinant theorem and its Padé consequences; give a direct natural-boundary argument for the Mahler product; and collect the transcendence and sharp irrationality-exponent results for Thue–Morse–Mahler numbers. A final section maps the paper results to existing Lean modules and proposes a dependency-aware formalization order for the additional material.

Every elementary identity is proved in the text. Results whose known proofs require deeper symbolic dynamics, Mahler’s method, or determinant theory are explicitly identified as literature theorems. No statement is described as formally verified unless a corresponding Lean declaration or module is named.

Contents

1	Scope, conventions, and the formula being complemented	4
1.1	The original Pascal-row logarithm	4
1.2	A compact formula index	5
2	Binary recursion, blocks, automata, and uniqueness	5
2.1	The two-scale recurrence	5
2.2	Concatenating more than one bit	6
2.3	Complement and permutation symmetries	6
2.4	The Boolean group law	6

3	Digit, floor, fractional-part, and trigonometric formulae	7
3.1	Extracting digits arithmetically	7
3.2	Fractional parts	7
3.3	Finite sign products	7
4	Two-adic valuations, central binomial coefficients, Catalan numbers, and carries	8
4.1	Legendre’s formula as a parity extractor	8
4.2	Catalan parity and the ruler sequence	8
4.3	Kummer’s carry cocycle	8
5	Pascal support, sparse columns, residue formulae, and Boolean inversion	9
5.1	Lucas support is a Boolean lattice	9
5.2	Power-of-two columns recover the digits	9
5.3	Removing the logarithm modulo three	10
5.4	A signed row-statistic family	10
5.5	An r -nomial logarithm family	11
5.6	Boolean Möbius inversion	11
6	Odious and evil enumerators	11
7	Lacunary generating products and their coefficient formulae	12
7.1	The finite mother product	12
7.2	The Mahler product	13
7.3	The logarithm reveals the ruler sequence	13
7.4	Partition and Bell-polynomial formulae	14
7.5	A lower-Hessenberg determinant	14
7.6	The bit series over characteristic two	15
7.7	An integer algebraic lift	15
7.8	A direct natural-boundary argument	16
8	Finite differences, Prouhet cancellation, and exact moments	16
8.1	A master mixed-difference identity	16
8.2	Prouhet annihilation and sharp extraction	17
8.3	The finite exponential transform	17
8.4	All higher dyadic power sums	18
8.5	Sparse Prouhet identities on one Pascal row	19
9	Exact prefix sums, discrepancy, and the Fabius connection	19
9.1	The first prefix is already exact	19
9.2	Iterated prefixes and Prouhet vanishing	20
9.3	Why the sequence appears in the Fabius theory	20
10	Ordinary Fourier transforms, Walsh characters, and Riesz products	21
10.1	The trigonometric factorization	21
10.2	The exact discrete Fourier transform	21
10.3	The Walsh transform is a delta function	21
10.4	Finite Riesz products	22
10.5	Autocorrelation recursion	22

11 Dirichlet transforms, digit constants, and infinite products	23
11.1 An entire shifted Dirichlet series	23
11.2 Thue–Morse digit constants in every base	23
11.3 The Woods–Robbins product	24
11.4 A useful product functional equation	24
12 Rarefied sums and Newman’s phenomenon	24
12.1 A root-of-unity filter at dyadic endpoints	24
12.2 Complete evaluation modulo three	25
12.3 The global Newman theorem	25
13 Combinatorics and dynamics of the infinite word	25
13.1 Uniform recurrence and aperiodicity	26
13.2 Complement and reversal closure	26
13.3 Overlap avoidance and critical exponent	26
13.4 Linear factor complexity	26
14 Hankel determinants, Mahler values, and Diophantine consequences	26
14.1 Hankel determinants built from the signs	26
14.2 Transcendence of the function and of algebraic values	27
15 The base-B root-of-unity family	27
16 Formula selection and a Lean formalization roadmap	28
16.1 Which representation is useful where?	28
16.2 Existing formalized spine	29
16.3 Suggested dependency order for the new formulae	29
16.4 Executable checks	30
17 Worked examples and consistency checks	31
17.1 One index through ten formulae	31
17.2 A complete four-bit block	31
17.3 A carry-corrected addition	32
18 Concluding synthesis	32
A The first 64 terms	33
B A compact identity map	33
C Consolidation provenance	34

1 Scope, conventions, and the formula being complemented

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Write the binary expansion of n as

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\}, \quad s_2(n) = \sum_{j \geq 0} b_j(n). \quad (1)$$

Only finitely many digits are nonzero. The two standard normalizations of the Thue–Morse sequence are

$$\tau(n) = s_2(n) \bmod 2 \in \{0, 1\}, \quad \varepsilon_n = (-1)^{s_2(n)} = 1 - 2\tau(n) \in \{1, -1\}. \quad (2)$$

Thus

$$\tau(n) = \frac{1 - \varepsilon_n}{2}, \quad \varepsilon_n = 1 - 2\tau(n). \quad (3)$$

The signed normalization is the natural one for products, finite differences, and Fourier analysis; the bit normalization is the natural one for substitutions and digit expansions. Keeping both visible prevents a large class of otherwise easy sign mistakes.

For bitwise exclusive-or we write $a \oplus b$, and $k \preccurlyeq_2 n$ means that every one-bit of k is also a one-bit of n . The symbol $[x]_m$ denotes the least nonnegative residue of an integer x modulo m . Empty sums are zero and empty products are one.

1.1 The original Pascal-row logarithm

The formula that motivated the drafts hides the binary expansion inside one row of Pascal’s triangle. Put

$$\mathcal{X}(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}}. \quad (4)$$

If $\nu(n)$ denotes the number of odd entries in row n , then each odd entry contributes -1 and each even entry contributes $+1$, so

$$\mathcal{X}(n) = n + 1 - 2\nu(n). \quad (5)$$

Lucas’s theorem gives $\nu(n) = 2^{s_2(n)}$, and hence

$$n + 1 - \mathcal{X}(n) = 2^{s_2(n)+1}. \quad (6)$$

The logarithm in the resulting formula is therefore exact, not an approximation:

$$\tau(n) = \frac{1 + (-1)^{\log_2 \left(n+1 - \sum_{k=0}^n (-1)^{\binom{n}{k}} \right)}}{2}. \quad (7)$$

The purpose of the atlas is not to restate (7) many times, but to show that it is one manifestation of a single parity character appearing in several apparently unrelated structures.

Proof-status boundary. The repository already formalizes the binary recursion, the finite block product, Prouhet cancellation and its sharp first surviving moment, iterated prefix sums, and the binomial–logarithm formula. The additional valuation, Pascal-residue, determinant, rarefied-sum, spectral, and analytic formulae are proved or cited here as ordinary mathematics; they are not silently promoted to existing Lean theorems.

1.2 A compact formula index

The following table previews representative identities. Their assumptions, proofs, and interrelations are developed later.

Viewpoint	Representative identity
Binary recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n.$
Block concatenation	$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r$ for $0 \leq r < 2^m.$
Boolean character	$\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b.$
Digit product	$\varepsilon_n = \prod_{j \geq 0} (1 - 2b_j(n)).$
Floor sum	$\varepsilon_n = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor}.$
Factorial valuation	$\varepsilon_n = (-1)^{n - v_2(n!)}.$
Central binomial	$\varepsilon_n = (-1)^{v_2 \binom{2n}{n}}.$
Carry cocycle	$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{v_2 \binom{a+b}{a}}.$
Sparse Pascal columns	$\varepsilon_n = (-1)^{\sum_j \binom{n}{2^j}}.$
Pascal support	$\nu(n) = 2^{s_2(n)},$ hence $\varepsilon_n = (-1)^{\log_2 \nu(n)}.$
Log-free Pascal form	$\tau(n) = [\nu(n)]_3 - 1.$
Finite product	$\sum_{n < 2^m} \varepsilon_n z^n = \prod_{j < m} (1 - z^{2^j}).$
Mahler equation	$E(z) = (1 - z)E(z^2).$
Ruler convolution	$n\varepsilon_n = -\sum_{k=1}^n (2^{v_2(k)+1} - 1)\varepsilon_{n-k}.$
Prouhet cancellation	$\sum_{n < 2^m} \varepsilon_n p(n) = 0$ for $\deg p < m.$
Sharp moment	$\sum_{n < 2^m} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}.$
Prefix discrepancy	$\sum_{n < N} \varepsilon_n = 0$ for even N , and $\varepsilon_{(N-1)/2}$ for odd $N.$
Walsh spectrum	The level- m block is one Boolean frequency.
Riesz product	$2^{-m} P_m(e^{2\pi i x}) ^2 = \prod_{j < m} (1 - \cos(2\pi 2^j x)).$
Dirichlet transform	$\sum_{n \geq 0} \varepsilon_n / (n+1)^s$ has an entire continuation.
Rarefied sum	Residue-class sums at 2^m are explicit powers of 3.

2 Binary recursion, blocks, automata, and uniqueness

2.1 The two-scale recurrence

Theorem 2.1 (Binary recursion and uniqueness). *The signed and zero–one sequences satisfy*

$$\varepsilon_0 = 1, \quad \varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n, \quad (8)$$

$$\tau(0) = 0, \quad \tau(2n) = \tau(n), \quad \tau(2n+1) = 1 - \tau(n). \quad (9)$$

Either row, with its initial value, uniquely determines its sequence.

Proof. Multiplication by two appends a zero binary digit, while passing from $2n$ to $2n+1$ replaces that terminal zero by one. Thus $s_2(2n) = s_2(n)$ and $s_2(2n+1) = s_2(n) + 1$, proving the recurrences. Every positive integer is $2q$ or $2q+1$ with $q < n$, so strong induction reduces any value to the initial value at zero. $\square$

The infinite zero–one word

$$\mathbf{t} = 0110100110010110 \dots$$

is the fixed point beginning in zero of the length-two morphism

$$\mu(0) = 01, \quad \mu(1) = 10. \quad (10)$$

In signed form, every block is followed by its negative. If $\varepsilon^{(m)} = (\varepsilon_0, \dots, \varepsilon_{2^m-1})$, then

$$\varepsilon^{(m+1)} = (\varepsilon^{(m)}, -\varepsilon^{(m)}), \quad \varepsilon^{(m)} = (1, -1)^{\otimes m}. \quad (11)$$

2.2 Concatenating more than one bit

Theorem 2.2 (Dyadic block multiplicativity). *For $m, q \geq 0$ and $0 \leq r < 2^m$,*

$$s_2(2^m q + r) = s_2(q) + s_2(r), \quad (12)$$

$$\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r, \quad (13)$$

$$\tau(2^m q + r) = \tau(q) \oplus \tau(r). \quad (14)$$

Proof. The binary digits of q are shifted above the lowest m positions, while r occupies only those positions. Their supports are disjoint, so the digit sums add without carries. Taking parity gives the last two identities. $\square$

For each fixed m, r , the subsequence $(\varepsilon_{2^m q + r})_{q \geq 0}$ is therefore either the original sequence or its negative. Consequently the signed 2-kernel has only two members; in the bit normalization the kernel consists of τ and $1 - \tau$. This is the two-state automaton content of (8).

2.3 Complement and permutation symmetries

Proposition 2.3 (Reflection in a dyadic block). *For $0 \leq r < 2^m$,*

$$\varepsilon_{2^m-1-r} = (-1)^m \varepsilon_r, \quad \tau(2^m - 1 - r) = \tau(r) \oplus (m \bmod 2). \quad (15)$$

More generally, permuting the first m binary positions leaves ε_r unchanged.

Proof. Within exactly m positions, $2^m - 1 - r$ is the bitwise complement of r , so its weight is $m - s_2(r)$. A permutation of positions does not change their sum. $\square$

Thus a dyadic word is a palindrome when m is even and an anti-palindrome when m is odd. This elementary identity later explains reciprocity of the finite generating polynomial and reversal closure of the factor language.

2.4 The Boolean group law

Theorem 2.4 (XOR character). *For all $a, b \geq 0$,*

$$\varepsilon_{a \oplus b} = \varepsilon_a \varepsilon_b, \quad \tau(a \oplus b) = \tau(a) \oplus \tau(b). \quad (16)$$

Proof. At every bit, exclusive-or is addition in $\mathbb{F}_2$. Hence $s_2(a \oplus b) \equiv s_2(a) + s_2(b) \pmod{2}$, and both identities follow. $\square$

This is the cleanest structural description of the signed sequence: ε is a one-dimensional character of the countable Boolean group of finite binary strings. Ordinary addition is more complicated only because it introduces carries; Section 4 will identify the exact correction factor.

3 Digit, floor, fractional-part, and trigonometric formulae

3.1 Extracting digits arithmetically

Lemma 3.1 (Floor extraction). *For every $j \geq 0$,*

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor. \quad (17)$$

Consequently

$$\varepsilon_n = \prod_{j \geq 0} \left(1 - 2 \left(\left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor \right) \right), \quad (18)$$

where all but finitely many factors are one.

Proof. The integer $\lfloor n/2^j \rfloor$ has least significant bit $b_j(n)$; subtracting twice its quotient by two extracts that bit. Since $(-1)^b = 1 - 2b$ for $b \in \{0, 1\}$, the product is $(-1)^{\sum_j b_j(n)}$. $\square$

Summing (17) with weights is not the only floor representation. The finite identity

$$s_2(n) = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor \quad (19)$$

is the base-two instance of Legendre’s digit-sum formula. It gives

$$\boxed{\varepsilon_n = (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor}}. \quad (20)$$

3.2 Fractional parts

Because $\sum_{j \geq 1} n/2^j = n$, subtracting the floor sum in (19) gives an exact infinite formula:

$$\boxed{s_2(n) = \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\}}. \quad (21)$$

The series is not effectively finite, but its tail is geometric once $2^j > n$. Thus

$$\varepsilon_n = \exp \left(\pi i \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} \right). \quad (22)$$

Although the exponent in (22) is presented as an infinite sum, (21) proves that it is the integer $s_2(n)$.

3.3 Finite sign products

For a nonintegral real x , $\operatorname{sgn}(\sin \pi x) = (-1)^{\lfloor x \rfloor}$. Since $\lfloor (n + 1/2)/2^j \rfloor = \lfloor n/2^j \rfloor$, summing the floor exponents yields $2n - s_2(n)$. Therefore

$$\boxed{\varepsilon_n = \prod_{j \geq 0} \operatorname{sgn} \sin \left(\frac{\pi(n + 1/2)}{2^j} \right)}. \quad (23)$$

Once $2^j > n + 1/2$, the sine argument lies in $(0, \pi)$, so the factor is $+1$. An equivalent cosine form is

$$\boxed{\varepsilon_n = \prod_{j \geq 0} \cos \left(\pi \left\lfloor \frac{n}{2^j} \right\rfloor \right)}. \quad (24)$$

These products are exact square-wave encodings; they do not smooth the sequence. The half-unit shift in (23) is essential because it prevents zero sine factors at integer arguments.

4 Two-adic valuations, central binomial coefficients, Catalan numbers, and carries

4.1 Legendre’s formula as a parity extractor

For a positive integer M , let $v_2(M)$ be the exponent of two in its prime factorization. Legendre’s formula gives

$$v_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - s_2(n). \quad (25)$$

Hence

$$\boxed{\varepsilon_n = (-1)^{n-v_2(n!)}, \quad \tau(n) = [n - v_2(n!)]_2.} \quad (26)$$

Applying (25) to $(2n)!/(n!)^2$ collapses the whole digit sum into one central binomial coefficient:

$$v_2\binom{2n}{n} = 2s_2(n) - s_2(2n) = s_2(n). \quad (27)$$

Thus

$$\boxed{\varepsilon_n = (-1)^{v_2\binom{2n}{n}}, \quad \tau(n) = \sin^2\left(\frac{\pi}{2} v_2\binom{2n}{n}\right).} \quad (28)$$

This is arguably the most economical arithmetic replacement for the row-wide binomial–logarithm formula: one valuation of one integer contains exactly $s_2(n)$.

4.2 Catalan parity and the ruler sequence

Let $C_n = (n+1)^{-1}\binom{2n}{n}$ be the n -th Catalan number. From (27),

$$v_2(C_n) = s_2(n) - v_2(n+1) = s_2(n+1) - 1. \quad (29)$$

Consequently

$$\boxed{\varepsilon_n = (-1)^{v_2(C_n)+v_2(n+1)}, \quad \varepsilon_{n+1} = -(-1)^{v_2(C_n)}.} \quad (30)$$

The second formula says that the parity of a Catalan valuation is the shifted complement of Thue–Morse.

Incrementing n flips its trailing ones to zero and the next zero to one. The number of trailing ones is $v_2(n+1)$, so

$$s_2(n+1) = s_2(n) + 1 - v_2(n+1). \quad (31)$$

Exponentiating gives a sequential recurrence with no explicit digit extraction:

$$\boxed{\varepsilon_{n+1} = -(-1)^{v_2(n+1)}\varepsilon_n.} \quad (32)$$

Thus the parity of the classical ruler function is the multiplicative first difference of Thue–Morse.

4.3 Kummer’s carry cocycle

Kummer’s theorem, in its base-two digit-sum form, states

$$v_2\binom{a+b}{a} = s_2(a) + s_2(b) - s_2(a+b). \quad (33)$$

The valuation is the number of binary carries, counted with propagation. Rearranging parity in (33) gives the exact obstruction to multiplicativity under ordinary addition.

Theorem 4.1 (Carry-corrected addition). *For all $a, b \geq 0$,*

$$\boxed{\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{v_2 \binom{a+b}{a}}.} \quad (34)$$

Equivalently,

$$\tau(a+b) = \tau(a) \oplus \tau(b) \oplus \left[v_2 \binom{a+b}{a} \right]_2. \quad (35)$$

Proof. Equation (33) rewrites $s_2(a+b)$ as $s_2(a) + s_2(b) - v_2 \binom{a+b}{a}$. Raise (-1) to both sides; subtraction and addition coincide modulo two. $\square$

In particular, if $a \& b = 0$, then addition is carry-free and

$$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b. \quad (36)$$

The dyadic block formula is the most important special case. More generally, if $N = a_1 + \dots + a_r$, then

$$\varepsilon_N = \left(\prod_{j=1}^r \varepsilon_{a_j} \right) (-1)^{v_2 \binom{N}{a_1, \dots, a_r}}, \quad (37)$$

proved by applying Legendre’s formula to the multinomial coefficient.

5 Pascal support, sparse columns, residue formulae, and Boolean inversion

5.1 Lucas support is a Boolean lattice

Lucas’s theorem modulo two says

$$\binom{n}{k} \equiv \prod_{j \geq 0} \binom{b_j(n)}{b_j(k)} \pmod{2}. \quad (38)$$

Every factor is one exactly when $b_j(k) \leq b_j(n)$. Hence

$$\binom{n}{k} \text{ is odd} \iff k \preceq_2 n. \quad (39)$$

The odd positions in row n are therefore literally the Boolean lattice on the one-bits of n . Its support polynomial factors as

$$Q_n(z) = \sum_{k=0}^n \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k = \prod_{j: b_j(n)=1} (1 + z^{2^j}). \quad (40)$$

Setting $z = 1$ proves Glaisher’s count

$$\boxed{\nu(n) = 2^{s_2(n)}.} \quad (41)$$

5.2 Power-of-two columns recover the digits

Taking $k = 2^j$ in Lucas’s theorem gives

$$\binom{n}{2^j} \equiv b_j(n) \pmod{2}. \quad (42)$$

Thus only logarithmically many Pascal entries are needed to reconstruct the parity bit:

$$\varepsilon_n = (-1)^{\sum_{j \geq 0} \binom{n}{2^j}} = \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right). \quad (43)$$

Terms with $2^j > n$ are zero. Formula (43) is a direct sparse companion to (7): it reads the row at power-of-two columns instead of counting all of its odd entries.

5.3 Removing the logarithm modulo three

Because 2^r alternates between 1 and 2 modulo three, Glaisher’s count itself reveals the parity of $s_2(n)$ without a logarithm.

Theorem 5.1 (Log-free odd-count formula). *For every $n \geq 0$,*

$$\tau(n) = [\nu(n)]_3 - 1, \quad \varepsilon_n = 3 - 2[\nu(n)]_3. \quad (44)$$

The residue $[\nu(n)]_3$ is always 1 or 2.

Proof. By (41), $\nu(n) = 2^{s_2(n)}$. Its residue is one when the exponent is even and two when it is odd. $\square$

A version using exactly the signs from the original formula is also available. Define

$$S(n) = \sum_{k=1}^n (-1)^{\binom{n}{k}}, \quad S(0) = 0. \quad (45)$$

Since the omitted $k = 0$ term equals -1 , equations (5) and (41) give

$$S(n) = n + 2 - 2^{s_2(n)+1}. \quad (46)$$

Therefore

$$\tau(n) = [2n + S(n)]_3, \quad (47)$$

$$\tau(n) = \frac{4}{3} \sin^2\left(\frac{\pi}{3}(n - S(n))\right). \quad (48)$$

Indeed $2n + S(n)$ has residue zero or one according as $s_2(n)$ is even or odd, while $n - S(n)$ has residue zero or two.

5.4 A signed row-statistic family

For a parameter u , put

$$R_u(n) = \sum_{k=0}^n u^{\binom{n}{k} \bmod 2}. \quad (49)$$

Even entries contribute one and odd entries contribute u , so

$$R_u(n) = n + 1 + (u - 1)2^{s_2(n)}. \quad (50)$$

The original statistic is $R_{-1}(n) = \mathcal{X}(n)$; $R_0(n)$ counts even entries; and any fixed $u \neq 1$ recovers the odd count from

$$2^{s_2(n)} = \frac{R_u(n) - (n + 1)}{u - 1}. \quad (51)$$

This makes clear that the special-looking sign sum is one point of a simple affine family.

5.5 An r -nomial logarithm family

Expand $(x_1 + \cdots + x_r)^n$. A multinomial coefficient $\binom{n}{k_1, \dots, k_r}$ is odd precisely when the binary additions $k_1 + \cdots + k_r = n$ are carry-free. At each one-bit of n , exactly one of the r indices receives that bit, independently of all other one-bits. Hence:

Theorem 5.2 (Odd multinomial count). *The number $\nu_r(n)$ of odd coefficients in the full r -nomial expansion of degree n is*

$$\boxed{\nu_r(n) = r^{s_2(n)}}. \quad (52)$$

For every integer $r \geq 2$,

$$\varepsilon_n = (-1)^{\log_r \nu_r(n)}, \quad \tau(n) = [\log_r \nu_r(n)]_2. \quad (53)$$

The binomial row is the case $r = 2$. The family shows that the logarithm is not tied to Pascal's triangle: it measures the number of independent choices attached to the one-bits.

5.6 Boolean Möbius inversion

The interval $[k, n]$ in the submask poset is a Boolean lattice of rank $s_2(n) - s_2(k)$. Its Möbius function is

$$\mu(k, n) = (-1)^{s_2(n) - s_2(k)} = \varepsilon_n \varepsilon_k \quad (k \preceq_2 n). \quad (54)$$

Therefore:

Theorem 5.3 (Thue–Morse Möbius inversion). *For functions $F, G : \mathbb{N} \rightarrow A$ valued in an abelian group, the relations*

$$F(n) = \sum_{k \preceq_2 n} G(k) \quad (55)$$

are equivalent to

$$\boxed{G(n) = \varepsilon_n \sum_{k \preceq_2 n} \varepsilon_k F(k)}. \quad (56)$$

In matrix form, let $P_{n,k} = \mathbf{1}_{\binom{n}{k} \text{ odd}}$ for $0 \leq k \leq n$. Every finite lower-triangular truncation has inverse

$$(P^{-1})_{n,k} = \varepsilon_n \varepsilon_k P_{n,k}. \quad (57)$$

Thus the Thue–Morse signs are exactly the Möbius kernel of Pascal's triangle reduced to its parity support.

6 Odious and evil enumerators

An integer is called *evil* when its binary weight is even and *odious* when its binary weight is odd. Let $\mathcal{E}_n$ and $\mathcal{O}_n$ be the n -th evil and odious nonnegative integers, respectively, with both sequences indexed from zero.

Theorem 6.1 (Exact evil and odious enumerators). *For every $n \geq 0$,*

$$\boxed{\mathcal{E}_n = 2n + \tau(n), \quad \mathcal{O}_n = 2n + 1 - \tau(n)}. \quad (58)$$

Consequently

$$\mathcal{O}_n - \mathcal{E}_n = \varepsilon_n, \quad \mathcal{E}_n + \mathcal{O}_n = 4n + 1. \quad (59)$$

Proof. The pair $2n, 2n + 1$ has weights $s_2(n)$ and $s_2(n) + 1$, so it contains exactly one evil and one odious number. If $\tau(n) = 0$, the even member $2n$ is evil; if $\tau(n) = 1$, the odd member $2n + 1$ is evil. This gives both enumerators, and (59) follows from $\varepsilon_n = 1 - 2\tau(n)$. $\square$

The formulas turn the sequence into a displacement between two complementary enumerations. They also give immediate counting functions. Let

$$D(N) = \sum_{n=0}^{N-1} \varepsilon_n. \quad (60)$$

Then the numbers of evil and odious integers below N are

$$E(N) = \frac{N + D(N)}{2}, \quad O(N) = \frac{N - D(N)}{2}. \quad (61)$$

The exact prefix formula in Section 9 will show $|D(N)| \leq 1$. Thus every even initial interval contains exactly as many evil as odious integers, and every odd initial interval differs by exactly one.

7 Lacunary generating products and their coefficient formulae

7.1 The finite mother product

For $m \geq 0$, define the level- m block polynomial

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n. \quad (62)$$

Theorem 7.1 (Finite lacunary product). *For every $m \geq 0$,*

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (63)$$

In particular,

$$P_{m+1}(z) = (1 - z^{2^m})P_m(z). \quad (64)$$

Proof. Expanding the product amounts to choosing a subset $S \subseteq \{0, \dots, m-1\}$. It contributes $(-1)^{|S|} z^{\sum_{j \in S} 2^j}$. Binary representation gives a unique subset for each $0 \leq n < 2^m$, and the coefficient is $(-1)^{s_2(n)} = \varepsilon_n$. $\square$

The product is the common source of the finite-difference, Prouhet, Fourier, and Riesz formulae below. It also displays exact reciprocity and cyclotomic multiplicities.

Proposition 7.2 (Reciprocity and cyclotomic structure). *Let $D = 2^m - 1$. Then*

$$z^D P_m(z^{-1}) = (-1)^m P_m(z), \quad (65)$$

and

$$P_m(z) = (-1)^m \prod_{q=0}^{m-1} \Phi_{2^q}(z)^{m-q}, \quad (66)$$

where $\Phi_1(z) = z - 1$. A root of unity of exact order 2^q , with $0 \leq q < m$, has multiplicity $m - q$.

Proof. For reciprocity,

$$z^D P_m(z^{-1}) = \prod_{j < m} (z^{2^j} - 1) = (-1)^m P_m(z).$$

Also $z^{2^j} - 1 = \prod_{q=0}^j \Phi_{2^q}(z)$. Since $1 - z^{2^j} = -(z^{2^j} - 1)$, multiplying over $j < m$ gives the sign and the exponent $m - q$ of each cyclotomic factor. $\square$

Coefficient comparison in (65) recovers the reflection formula (15). The zero of order m at $z = 1$ is the generating-function shadow of Prouhet cancellation.

7.2 The Mahler product

The stabilized series

$$E(z) = \sum_{n \geq 0} \varepsilon_n z^n \quad (67)$$

exists coefficientwise as a formal series. For $|z| < 1$, the product converges normally on compact subsets because $\sum_j |z|^{2^j} < \infty$. Passing to the limit in (63) gives

$$E(z) = \prod_{j \geq 0} (1 - z^{2^j}), \quad E(z) = (1 - z)E(z^2). \quad (68)$$

The second identity is the first-order Mahler functional equation obtained by removing the $j = 0$ factor and reindexing.

For the bit series

$$T(z) = \sum_{n \geq 0} \tau(n) z^n, \quad (69)$$

we have

$$T(z) = \frac{1}{2} \left(\frac{1}{1 - z} - E(z) \right), \quad (70)$$

$$T(z) = (1 - z)T(z^2) + \frac{z}{1 - z^2}. \quad (71)$$

The first identity is the generating form of $\varepsilon_n = 1 - 2\tau(n)$; the second can also be proved directly by separating even and odd indices and using (9).

7.3 The logarithm reveals the ruler sequence

Set

$$a_k = \sum_{2^j | k} 2^j = 1 + 2 + \dots + 2^{v_2(k)} = 2^{v_2(k)+1} - 1. \quad (72)$$

For $|z| < 1$, absolute convergence permits the rearrangement

$$\begin{aligned} \log E(z) &= \sum_{j \geq 0} \log(1 - z^{2^j}) = - \sum_{j \geq 0} \sum_{r \geq 1} \frac{z^{r 2^j}}{r} \\ &= - \sum_{k \geq 1} \frac{a_k}{k} z^k. \end{aligned} \quad (73)$$

Indeed, a term of degree k occurs for exactly the powers $2^j \mid k$, and its contribution is $-2^j/k$.

Logarithmic differentiation gives an unusual all-previous-terms recurrence.

Theorem 7.3 (Ruler-convolution recurrence). *With $a_k = 2^{v_2(k)+1} - 1$,*

$$n\varepsilon_n = - \sum_{k=1}^n a_k \varepsilon_{n-k} \quad (n \geq 1), \quad (74)$$

with $\varepsilon_0 = 1$.

Proof. Differentiate (73), multiply by $E(z) = \sum_{r \geq 0} \varepsilon_r z^r$, and compare the coefficient of z^{n-1} . $\square$

The weighted convolution on the right of (74) is therefore always exactly $\pm n$. Unlike the two-state recursion, this formula contains no digits of n ; all binary information is hidden in the valuations of the ordinary indices $1, \dots, n$.

7.4 Partition and Bell-polynomial formulae

Exponentiating (73) and collecting terms of total degree n gives a closed finite sum over integer partitions.

Theorem 7.4 (Partition formula). *For every $n \geq 0$,*

$$\varepsilon_n = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \prod_{j=1}^n \frac{1}{m_j!} \left(-\frac{a_j}{j} \right)^{m_j}, \quad a_j = 2^{v_2(j)+1} - 1. \quad (75)$$

At $n = 0$, the unique empty choice contributes one.

Proof. Write

$$E(z) = \prod_{j \geq 1} \exp\left(-\frac{a_j}{j} z^j\right) = \prod_{j \geq 1} \sum_{m_j \geq 0} \frac{1}{m_j!} \left(-\frac{a_j}{j}\right)^{m_j} z^{jm_j}.$$

The coefficient of z^n receives exactly the choices with $\sum_j jm_j = n$. $\square$

Define the complete exponential Bell polynomials by

$$\exp\left(\sum_{j \geq 1} x_j \frac{z^j}{j!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (76)$$

Then (73) yields the compact equivalent form

$$\varepsilon_n = \frac{1}{n!} \mathcal{B}_n(-0!a_1, -1!a_2, \dots, -(n-1)!a_n). \quad (77)$$

The rational-looking partition sum and Bell polynomial always collapse to ± 1 ; the cancellation is highly nonlocal and invisible term by term.

7.5 A lower-Hessenberg determinant

The same recurrence admits a determinant solution. For $n \geq 1$, let

$$H_n = \begin{pmatrix} a_1 & 1 & 0 & 0 & \cdots & 0 \\ a_2 & a_1 & 2 & 0 & \cdots & 0 \\ a_3 & a_2 & a_1 & 3 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & 0 \\ a_{n-1} & a_{n-2} & \cdots & a_2 & a_1 & n-1 \\ a_n & a_{n-1} & \cdots & a_3 & a_2 & a_1 \end{pmatrix}. \quad (78)$$

Theorem 7.5 (Hessenberg determinant formula). *For every $n \geq 1$,*

$$\boxed{\varepsilon_n = \frac{(-1)^n}{n!} \det H_n.} \quad (79)$$

In particular, $|\det H_n| = n!$.

Proof. Let $D_0 = 1$ and $D_n = \det H_n$. Expansion of a lower-Hessenberg determinant along the last row, or induction by continuant elimination, gives

$$D_n = \sum_{r=1}^n (-1)^{r+1} \frac{(n-1)!}{(n-r)!} a_r D_{n-r}. \quad (80)$$

Put $c_n = (-1)^n D_n / n!$. Multiplying (80) by $(-1)^n / n!$ gives

$$nc_n = - \sum_{r=1}^n a_r c_{n-r}, \quad c_0 = 1.$$

This is exactly (74), whose solution is unique by induction. $\square$

The determinant is structurally different from the Hankel determinants discussed in Section 14. Here the matrix is engineered from the logarithmic derivative and its determinant has the elementary magnitude $n!$; the classical Hankel matrix uses the Thue–Morse terms themselves and has a much subtler nonvanishing theorem.

7.6 The bit series over characteristic two

Reducing (71) coefficientwise modulo two, use $T(z^2) = T(z)^2$, $1 - z = 1 + z$, and $1 + z^2 = (1 + z)^2$. One obtains

$$\boxed{(1 + z)^3 T(z)^2 + (1 + z)^2 T(z) + z = 0 \quad \text{in } \mathbb{F}_2[[z]].} \quad (81)$$

This is an explicit algebraic certificate of automaticity. It is also a concrete instance of the Christol–Kamae–Mendès France–Rauzy theorem: over a finite field, a sequence is automatic exactly when its generating series is algebraic over the rational-function field [9].

7.7 An integer algebraic lift

A surprisingly different integer sequence has Thue–Morse parity. Define $c_0 = 0$, $c_1 = 1$, and for $n \geq 2$,

$$nc_n = (5 - 2n)c_{n-1} + 3(n-1)c_{n-2} + 1. \quad (82)$$

Its first terms are

$$0, 1, 1, 2, 1, 4, -2, 13, -23, 68, -164, 439, -1146, 3067, \dots$$

Theorem 7.6 (Algebraic integer lift). *Let $C(z) = \sum_{n \geq 0} c_n z^n$. Then*

$$C(z) = \frac{1}{2(1-z)} \left(\sqrt{\frac{1+3z}{1-z}} - 1 \right), \quad (83)$$

$$(1-z)^3 C(z)^2 + (1-z)^2 C(z) = z. \quad (84)$$

All c_n are integers and

$$\boxed{c_n \equiv \tau(n) \pmod{2}, \quad \varepsilon_n = (-1)^{c_n}.} \quad (85)$$

Proof. Multiplying (82) by z^{n-1} , summing, and using the initial conditions gives a first-order differential equation whose solution is (83); squaring gives (84). Conversely, because $C(0) = 0$, coefficient comparison in (84) determines each new coefficient from earlier integer coefficients, proving integrality and uniqueness.

Modulo two, (84) becomes exactly (81). That equation has a unique solution in $z\mathbb{F}_2[[z]]$: at each degree the linear term contains the new coefficient with coefficient one, while the square uses only earlier or lower-index data. Hence $C(z) \equiv T(z) \pmod{2}$. $\square$

This lift is not a disguised binary weight: its coefficients grow, change sign, and are nonautomatic over $\mathbb{Z}$. The parity survives because an algebraic equation over $\mathbb{Z}[[z]]$ reduces to the characteristic-two equation of the automatic sequence.

7.8 A direct natural-boundary argument

The unit circle is a natural boundary for E . One can invoke the Pólya–Carlson theorem, but the product gives a concrete proof. Every root of unity whose order is a power of two is a radial zero: if $\zeta^{2^J} = 1$, then for $0 < r < 1$ the tail contains $\prod_{j \geq J} (1 - r^{2^j})$, which tends to zero as $r \uparrow 1$. Such roots are dense on the unit circle. If E admitted analytic continuation through any open boundary arc, the continuation would vanish at a dense set in that arc and hence, by the identity theorem, would vanish identically in a neighborhood. This contradicts $E(0) = 1$. Therefore no boundary arc can be crossed.

8 Finite differences, Prouhet cancellation, and exact moments

8.1 A master mixed-difference identity

For $h \in \mathbb{R}$, let $T_h f(x) = f(x + h)$ and $\Delta_h = T_h - I$. Translation operators commute, and the coefficient of T_{nh} in $\prod_{j < m} (I - T_{2^j h})$ is ε_n . Hence:

Theorem 8.1 (Mixed finite-difference identity). *For every function on which the displayed translations are defined,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x + nh) = \prod_{j=0}^{m-1} (I - T_{2^j h}) f(x) = (-1)^m \Delta_h \Delta_{2h} \cdots \Delta_{2^{m-1}h} f(x). \quad (86)$$

For $h > 0$ and $f \in C^m$, iterating $\Delta_a f(x) = \int_0^a f'(x + u) du$ gives the integral form

$$\sum_{n < 2^m} \varepsilon_n f(x + nh) = (-1)^m \int_0^h \int_0^{2h} \cdots \int_0^{2^{m-1}h} f^{(m)}(x + u_0 + \cdots + u_{m-1}) du_{m-1} \cdots du_0. \quad (87)$$

This formula contains both exact cancellation and sign information. For example, if $x > 0$, $h > 0$, and $s > 0$, then

$$\sum_{n < 2^m} \frac{\varepsilon_n}{(x + nh)^s} = (s)_m \int_0^h \cdots \int_0^{2^{m-1}h} \frac{du_{m-1} \cdots du_0}{(x + u_0 + \cdots + u_{m-1})^{s+m}} > 0, \quad (88)$$

where $(s)_m = s(s+1) \cdots (s+m-1)$. The sign $(-1)^m$ in (87) cancels the sign of the m -th derivative of y^{-s} .

8.2 Prouhet annihilation and sharp extraction

If p is a polynomial of degree below m , then its m -th derivative vanishes. Equation (87) therefore gives Prouhet’s construction.

Theorem 8.2 (Prouhet annihilation). *For every polynomial p with $\deg p < m$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(n) = 0. \quad (89)$$

Equivalently, for $0 \leq r < m$,

$$\sum_{n < 2^m} \varepsilon_n n^r = 0. \quad (90)$$

Since $\varepsilon_n = +1$ on the evil class and -1 on the odious class, (90) says that the two classes below 2^m have equal sums of like powers through degree $m - 1$. At the next degree the cancellation is exact but no longer zero.

Theorem 8.3 (First surviving moment). *For every $m \geq 0$,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}. \quad (91)$$

More generally, if $p(x) = a_m x^m + \cdots + a_0$, then

$$\sum_{n < 2^m} \varepsilon_n p(n) = a_m (-1)^m m! 2^{\binom{m}{2}}. \quad (92)$$

Proof. Take $h = 1$ and $f(x) = x^m$ in (87). The derivative is the constant $m!$, and the rectangular integration volume is $1 \cdot 2 \cdots 2^{m-1} = 2^{\binom{m}{2}}$. Lower-degree terms vanish by Theorem 8.2. $\square$

This is stronger than a vanishing theorem: on the vector space of polynomials of degree at most m , a complete Thue–Morse block is an exact leading-coefficient functional.

8.3 The finite exponential transform

Define

$$F_m(t) = \sum_{n=0}^{2^m-1} \varepsilon_n e^{nt}. \quad (93)$$

Substituting $z = e^t$ in (63) gives

$$F_m(t) = \prod_{j=0}^{m-1} (1 - e^{2^j t}). \quad (94)$$

Using $1 - e^u = -2e^{u/2} \sinh(u/2)$ and $\sum_{j < m} 2^j = 2^m - 1$,

$$F_m(t) = (-2)^m e^{(2^m-1)t/2} \prod_{j=0}^{m-1} \sinh(2^{j-1}t). \quad (95)$$

After centering the block at $c_m = (2^m - 1)/2$,

$$\sum_{n < 2^m} \varepsilon_n e^{(n-c_m)t} = (-2)^m \prod_{j=0}^{m-1} \sinh(2^{j-1}t). \quad (96)$$

The zero of exact order m at the origin simultaneously proves (90) and (91).

8.4 All higher dyadic power sums

Let

$$M_{m,r} = \sum_{n=0}^{2^m-1} \varepsilon_n n^r. \quad (97)$$

Since $F_m(t) = \sum_{r \geq 0} M_{m,r} t^r / r!$, expanding each factor in (94) gives a finite formula valid at every order.

Theorem 8.4 (Composition formula for all moments). *For $r \geq 0$,*

$$M_{m,r} = (-1)^m r! \sum_{\substack{q_0, \dots, q_{m-1} \geq 1 \\ q_0 + \dots + q_{m-1} = r}} \frac{2^{\sum_{j=0}^{m-1} j q_j}}{q_0! \cdots q_{m-1}!}. \quad (98)$$

The sum is empty, and hence zero, when $r < m$.

Proof. In the j -th factor,

$$1 - e^{2^j t} = - \sum_{q_j \geq 1} \frac{2^{j q_j} t^{q_j}}{q_j!}.$$

Multiply the m series and compare the coefficient of t^r . □

There is also a compact Bell-polynomial form. Let B_{2r} be the Bernoulli numbers, with $B_2 = 1/6$, and define

$$x_1^{(m)} = \frac{2^m - 1}{2}, \quad (99)$$

$$x_{2r}^{(m)} = \frac{B_{2r}}{2r} \frac{4^r - 1}{4^r - 1} \quad (r \geq 1), \quad (100)$$

$$x_{2r+1}^{(m)} = 0 \quad (r \geq 1). \quad (101)$$

The expansion

$$\log \frac{\sinh y}{y} = \sum_{r \geq 1} \frac{2^{2r-1} B_{2r}}{r(2r)!} y^{2r}$$

turns (95) into

$$F_m(t) = (-1)^m 2^{\binom{m}{2}} t^m \exp \left(\sum_{k \geq 1} x_k^{(m)} \frac{t^k}{k!} \right). \quad (102)$$

Therefore:

Corollary 8.5 (Bell-polynomial moments). *For every $\ell \geq 0$,*

$$M_{m,m+\ell} = (-1)^m 2^{\binom{m}{2}} \frac{(m+\ell)!}{\ell!} \mathcal{B}_\ell \left(x_1^{(m)}, \dots, x_\ell^{(m)} \right). \quad (103)$$

For example,

$$M_{m,m+1} = (-1)^m 2^{\binom{m}{2}} (m+1)! \frac{2^m - 1}{2}, \quad (104)$$

$$M_{m,m+2} = (-1)^m 2^{\binom{m}{2}} (m+2)! \left(\frac{(2^m - 1)^2}{8} + \frac{4^m - 1}{72} \right). \quad (105)$$

Thus the familiar first surviving moment is only the first coefficient of a complete exact hierarchy.

8.5 Sparse Prouhet identities on one Pascal row

The usual Prouhet block is the full submask set of $2^m - 1$. Lucas’s criterion suggests replacing it by an arbitrary Pascal row. Suppose the one-bit positions of n are $a_1 < \dots < a_r$. Then

$$\sum_{k \preccurlyeq_2 n} \varepsilon_k z^k = \prod_{j=1}^r (1 - z^{2^{a_j}}). \quad (106)$$

The right side has a zero of exact order $r = s_2(n)$ at $z = 1$.

Theorem 8.6 (Sparse Pascal-row Prouhet theorem). *Let n have one-bit positions $a_1, \dots, a_r$. Then*

$$\sum_{k \preccurlyeq_2 n} \varepsilon_k k^d = 0 \quad (0 \leq d < r), \quad (107)$$

$$\sum_{k \preccurlyeq_2 n} \varepsilon_k k^r = (-1)^r r! 2^{a_1 + \dots + a_r}. \quad (108)$$

Equivalently, the odd support of row n of Pascal’s triangle, signed by Thue–Morse, annihilates every polynomial of degree below the number of one-bits of n .

Proof. Set $z = e^t$ in (106). Near zero,

$$\prod_{j=1}^r (1 - e^{2^{a_j} t}) = (-1)^r 2^{a_1 + \dots + a_r} t^r + \mathcal{O}(t^{r+1}).$$

Compare exponential generating coefficients. □

Example 8.7. For $n = 13 = (1101)_2$, the one-bit positions are 0, 2, 3. The odd positions in Pascal row 13 are the submasks 0, 1, 4, 5, 8, 9, 12, 13. Their Thue–Morse-signed sums of powers vanish through degree two, and

$$\sum_{k \preccurlyeq_2 13} \varepsilon_k k^3 = (-1)^3 3! 2^{0+2+3} = -192.$$

9 Exact prefix sums, discrepancy, and the Fabius connection

9.1 The first prefix is already exact

Define the half-open signed prefix sum

$$D(N) = \sum_{n=0}^{N-1} \varepsilon_n. \quad (109)$$

Pairing consecutive terms gives a complete formula.

Theorem 9.1 (Exact prefix discrepancy). *For every $N \geq 0$,*

$$D(N) = \begin{cases} 0, & N \text{ even,} \\ \varepsilon_{(N-1)/2}, & N \text{ odd.} \end{cases} \quad (110)$$

In particular, $|D(N)| \leq 1$.

Proof. For $N = 2M$, group the sum into pairs $\varepsilon_{2q} + \varepsilon_{2q+1} = \varepsilon_q - \varepsilon_q = 0$. For $N = 2M + 1$, the same pairs leave $\varepsilon_{2M} = \varepsilon_M$. □

Thus every interval has uniformly bounded discrepancy:

$$\left| \sum_{n=A}^{B-1} \varepsilon_n \right| = |D(B) - D(A)| \leq 2. \quad (111)$$

Every aligned dyadic block of positive length has sum zero:

$$\sum_{r=0}^{2^m-1} \varepsilon_{q2^m+r} = \varepsilon_q \sum_{r<2^m} \varepsilon_r = 0 \quad (m \geq 1). \quad (112)$$

This is stronger than asymptotic equidistribution; it is exact local balance at every dyadic scale.

9.2 Iterated prefixes and Prouhet vanishing

Let $(\mathcal{P}a)(N) = \sum_{0 \leq n < N} a(n)$, and let $\mathcal{P}^r$ denote r iterations. For $r \geq 1$, repeated summation gives

$$(\mathcal{P}^r a)(N) = \sum_{n=0}^{N-1} \binom{N-n-1}{r-1} a(n). \quad (113)$$

For fixed $N = 2^m$, the binomial weight is a polynomial in n of degree $r-1$. Consequently

$$(\mathcal{P}^r \varepsilon)(2^m) = 0 \quad (1 \leq r \leq m). \quad (114)$$

Different inclusive/exclusive endpoint conventions shift this formula by one and account for several superficially different signs in the literature and in computational experiments. The invariant content is the same: an m -bit block kills all prefix kernels of polynomial degree below m .

9.3 Why the sequence appears in the Fabius theory

Let up denote Rvachev’s even compactly supported up-function. In the main Fabius–Rvachev development, the signed global extension is represented by the locally finite translate sum

$$\boxed{\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1).} \quad (115)$$

Because up is supported on $[-1, 1]$, only finitely many terms contribute at any fixed x . The alternating binary block structure is exactly what converts the refinement law for up into the pure global dilation equation

$$\mathcal{F}'(x) = 2\mathcal{F}(2x). \quad (116)$$

The same signs govern exact dyadic evaluations of the bounded Fabius distribution function. Finite spline and moment formulae produce sums over complete or truncated dyadic blocks; [Theorem 8.2](#) removes all polynomial terms below the critical degree, while [Theorem 8.3](#) supplies the first nonzero contribution with its exact power of two and factorial. The block product (63) is the coefficient form of that cancellation. Thus the Thue–Morse layer is not decorative: it is the arithmetic mechanism that makes the dyadic formulas collapse.

The relevant formal modules include `ThueMorseGenerating.lean`, `ThueMorseExponential.lean`, `ThueMorsePrefix.lean`, `ThueMorseApproximation.lean`, `PaperKfoldThueMorse.lean`, and the exact dyadic modules listed in [Section 16](#). Formula (7) is formalized separately in `ThueMorseBinomialLog.lean`.

10 Ordinary Fourier transforms, Walsh characters, and Riesz products

10.1 The trigonometric factorization

On the unit circle, (63) becomes a finite sine product. Since $1 - e^{iu} = -2ie^{iu/2} \sin(u/2)$,

$$\sum_{n < 2^m} \varepsilon_n e^{in\theta} = (-2i)^m e^{i(2^m-1)\theta/2} \prod_{j=0}^{m-1} \sin(2^{j-1}\theta). \quad (117)$$

The phase records the center of the block; the magnitude factors scale by scale:

$$\left| \sum_{n < 2^m} \varepsilon_n e^{in\theta} \right| = 2^m \prod_{j=0}^{m-1} |\sin(2^{j-1}\theta)|. \quad (118)$$

This is the oscillatory counterpart of the centered hyperbolic-sine formula (96).

10.2 The exact discrete Fourier transform

Let $N = 2^m$, and use the convention

$$\widehat{\varepsilon}_m(k) = \sum_{n=0}^{N-1} \varepsilon_n e^{-2\pi i kn/N} \quad (0 \leq k < N). \quad (119)$$

Evaluation of the block polynomial gives

$$\widehat{\varepsilon}_m(k) = \prod_{j=0}^{m-1} \left(1 - e^{-2\pi i k 2^j / N} \right). \quad (120)$$

If $k \neq 0$ is even, one factor vanishes: writing $k = 2^v u$ with u odd, take $j = m - v$. Hence, for $m \geq 1$, the ordinary spectrum is supported on odd frequencies:

$$\widehat{\varepsilon}_m(k) = 0 \quad (k \text{ even}). \quad (121)$$

For odd k , every factor is nonzero, and Fourier inversion gives a finite cyclotomic formula for each individual sign:

$$\varepsilon_n = \frac{1}{2^m} \sum_{\substack{0 \leq k < 2^m \\ k \text{ odd}}} \left[\prod_{j=0}^{m-1} \left(1 - e^{-2\pi i k 2^j / 2^m} \right) \right] e^{2\pi i kn / 2^m}, \quad 0 \leq n < 2^m. \quad (122)$$

This is computationally inferior to the digit formula, but it is the natural exact representation for cyclic spectral questions.

10.3 The Walsh transform is a delta function

Identify $0 \leq n < 2^m$ with its bit vector $\mathbf{b}(n) = (b_0(n), \dots, b_{m-1}(n)) \in \mathbb{F}_2^m$. For $\mathbf{a} \in \mathbb{F}_2^m$, let

$$\chi_{\mathbf{a}}(\mathbf{b}) = (-1)^{\mathbf{a} \cdot \mathbf{b}}. \quad (123)$$

The Thue–Morse sign is the top-order character $\chi_{\mathbf{1}}$, where $\mathbf{1} = (1, \dots, 1)$.

Theorem 10.1 (One-point Walsh spectrum). *For $0 \leq r < 2^m$,*

$$\sum_{n=0}^{2^m-1} (-1)^{s_2(n)+s_2(r \& n)} = \begin{cases} 2^m, & r = 2^m - 1, \\ 0, & r \neq 2^m - 1. \end{cases} \quad (124)$$

Proof. The summand is the product $\chi_1(\mathbf{b}(n))\chi_{\mathbf{b}(r)}(\mathbf{b}(n))$, hence the character $\chi_{1+\mathbf{b}(r)}$. A nontrivial character sums to zero over $\mathbb{F}_2^m$, while the trivial character sums to 2^m . $\square$

Thus Thue–Morse is spectrally trivial for XOR: it is one Walsh frequency. The rich ordinary Fourier behavior comes entirely from replacing Boolean addition by cyclic addition.

10.4 Finite Riesz products

Normalize the ordinary Fourier intensity by

$$R_m(x) = 2^{-m} |P_m(e^{2\pi i x})|^2. \quad (125)$$

Since $|1 - e^{iu}|^2 = 2(1 - \cos u)$,

$$R_m(x) = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)). \quad (126)$$

Parseval’s identity, or the constant coefficient of the product, gives

$$\int_0^1 R_m(x) dx = 1. \quad (127)$$

Thus $R_m(x) dx$ is a probability measure. It does not converge to a smooth density; its weak limit is the classical singular continuous Thue–Morse spectral measure [5, 19]. Formula (126) is the exact finite object from which that measure is built.

10.5 Autocorrelation recursion

Define the limiting autocorrelation, whose existence follows from unique ergodicity of the primitive substitution system, by

$$\eta(r) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \varepsilon_n \varepsilon_{n+r} \quad (r \geq 0). \quad (128)$$

Separating even and odd indices and using (8) gives

$$\eta(0) = 1, \quad \eta(2r) = \eta(r), \quad \eta(2r+1) = -\frac{\eta(r) + \eta(r+1)}{2}. \quad (129)$$

For example, the odd recurrence at $r = 0$ gives $\eta(1) = -\frac{1}{2}(1 + \eta(1))$, hence

$$\eta(1) = -\frac{1}{3}, \quad \eta(2) = -\frac{1}{3}, \quad \eta(3) = \frac{1}{3}. \quad (130)$$

The numbers $\eta(r)$ are the Fourier coefficients of the limiting Riesz-product measure. The recursion is elementary; the theorem that the measure is purely singular continuous is a deeper spectral result.

11 Dirichlet transforms, digit constants, and infinite products

11.1 An entire shifted Dirichlet series

Consider

$$\mathcal{D}(s) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n+1)^s}. \quad (131)$$

The bounded prefix sums from [Theorem 9.1](#) imply convergence for $\Re s > 0$ by summation by parts, and absolute convergence for $\Re s > 1$. In the latter half-plane, the Mellin formula for $(n+1)^{-s}$ gives

$$\Gamma(s)\mathcal{D}(s) = \int_0^\infty t^{s-1} e^{-t} E(e^{-t}) dt. \quad (132)$$

Theorem 11.1 (Entire continuation). *The function $\mathcal{D}(s)$ extends to an entire function on $\mathbb{C}$. At every nonnegative integer r ,*

$$\boxed{\mathcal{D}(-r) = 0.} \quad (133)$$

For real $s > 0$, $\mathcal{D}(s) > 0$.

Proof. As $t \rightarrow \infty$, the factor e^{-t} gives exponential decay. As $t \downarrow 0$, choose J with $2^J t \leq 1 < 2^{J+1} t$. For $j \leq J$, $1 - e^{-2^j t} \leq 2^j t$, so

$$0 < E(e^{-t}) \leq t^{J+1} 2^{J(J+1)/2}.$$

Since $J \asymp \log(1/t)/\log 2$, this bound is smaller than every fixed power of t . Therefore the Mellin integral in (132) defines an entire function of s . Multiplication by the entire function $1/\Gamma(s)$ gives the entire continuation of $\mathcal{D}$; the zeros of $1/\Gamma$ at $0, -1, -2, \dots$ give (133). Positivity follows because every factor of $E(e^{-t})$ is positive for $t > 0$. $\square$

The vanishing values are the analytic continuation of Prouhet cancellation: formally, $\mathcal{D}(-r)$ would be a divergent sum of $\varepsilon_n(n+1)^r$, and the continuation assigns zero to every nonnegative polynomial degree.

11.2 Thue–Morse digit constants in every base

For an integer $b \geq 2$, define the base- b number with Thue–Morse digits

$$\kappa_b = \sum_{n \geq 0} \frac{\tau(n)}{b^{n+1}}. \quad (134)$$

Evaluating (70) at $z = 1/b$ gives

$$\boxed{\kappa_b = \frac{1}{2(b-1)} - \frac{1}{2b} \prod_{j \geq 0} \left(1 - b^{-2^j}\right).} \quad (135)$$

For $b = 2$, this is the Prouhet–Thue–Morse constant

$$\kappa_2 = 0.0110100110010110\dots_2 = \frac{1}{2} - \frac{1}{4} \prod_{j \geq 0} \left(1 - 2^{-2^j}\right). \quad (136)$$

The digit expansion and the lacunary product are therefore two evaluations of the same Mahler function.

11.3 The Woods–Robbins product

A different global fingerprint uses the signs as exponents rather than coefficients. The classical Woods–Robbins identity is

$$\prod_{n=0}^{\infty} \left(\frac{2n+1}{2n+2} \right)^{\varepsilon_n} = \frac{1}{\sqrt{2}}. \quad (137)$$

Convergence follows from the bounded prefix sums applied to the logarithms. The evaluation is obtained by separating even and odd indices, applying $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$, and simplifying the resulting functional identity; see Woods and Robbins and the systematic extensions in [24, 22, 4]. Many related products reduce to gamma values, but (137) remains the cleanest example.

11.4 A useful product functional equation

For parameters for which the product converges, define

$$\mathcal{P}(a, b) = \prod_{n \geq 0} \left(\frac{n+a}{n+b} \right)^{\varepsilon_n}. \quad (138)$$

Splitting even and odd indices gives the exact renormalization law

$$\mathcal{P}(a, b) = \mathcal{P}\left(\frac{a}{2}, \frac{b}{2}\right) \mathcal{P}\left(\frac{b+1}{2}, \frac{a+1}{2}\right). \quad (139)$$

This identity is the multiplicative analogue of the Mahler equation. Special parameter cycles are the mechanism behind closed evaluations such as (137).

12 Rarefied sums and Newman’s phenomenon

The bounded ordinary prefix discrepancy does not imply cancellation after restricting to an arithmetic progression. The most famous counterexample is the progression of multiples of three, where one sign has a persistent excess.

12.1 A root-of-unity filter at dyadic endpoints

For $q \geq 2$, $0 \leq r < q$, and $m \geq 0$, define

$$A_{q,r}(m) = \sum_{\substack{0 \leq n < 2^m \\ n \equiv r \pmod{q}}} \varepsilon_n. \quad (140)$$

Let $\zeta_q = e^{2\pi i/q}$. The usual root-of-unity filter and (63) give the exact finite formula

$$A_{q,r}(m) = \frac{1}{q} \sum_{\ell=0}^{q-1} \zeta_q^{-r\ell} \prod_{j=0}^{m-1} \left(1 - \zeta_q^{\ell 2^j} \right). \quad (141)$$

For $m \geq 1$, the $\ell = 0$ term vanishes. Formula (141) reduces the problem to the orbit of multiplication by two modulo q ; for odd q , that orbit is periodic.

12.2 Complete evaluation modulo three

Let $\omega = e^{2\pi i/3}$. The powers of two alternate modulo three, and

$$(1 - \omega)(1 - \omega^2) = 3. \quad (142)$$

This makes the rarefied sums completely explicit.

Theorem 12.1 (Dyadic residue-class sums modulo three). *For $u \geq 1$,*

$$(A_{3,0}(2u), A_{3,1}(2u), A_{3,2}(2u)) = (2 \cdot 3^{u-1}, -3^{u-1}, -3^{u-1}). \quad (143)$$

For $u \geq 0$,

$$(A_{3,0}(2u+1), A_{3,1}(2u+1), A_{3,2}(2u+1)) = (3^u, -3^u, 0). \quad (144)$$

Proof. For $m = 2u$, the factors in $P_m(\omega)$ occur in u pairs, each equal to three by (142); hence $P_{2u}(\omega) = P_{2u}(\omega^2) = 3^u$. Substitute these values into (141) and use $1 + \omega + \omega^2 = 0$.

For $m = 2u + 1$, one extra factor remains:

$$P_{2u+1}(\omega) = 3^u(1 - \omega), \quad P_{2u+1}(\omega^2) = 3^u(1 - \omega^2).$$

The three filtered combinations are then $3^u, -3^u, 0$. □

Along dyadic endpoints, the excess on multiples of three therefore grows with exponent

$$\alpha = \log_4 3 : \quad A_{3,0}(m) \asymp (2^m)^\alpha. \quad (145)$$

More precisely, the constant is $2/3$ when m is even and $1/\sqrt{3}$ when m is odd after factoring out $(2^m)^\alpha$.

12.3 The global Newman theorem

For an arbitrary endpoint, put

$$N_3(X) = \sum_{\substack{0 \leq n < X \\ 3|n}} \varepsilon_n. \quad (146)$$

Newman's theorem states that

$$N_3(X) > 0 \quad (X \geq 1). \quad (147)$$

Thus among multiples of three below every positive cutoff, those with even binary weight outnumber those with odd binary weight. This is not a generic balance phenomenon: it is a rigid resonance between multiplication by two modulo three and the sign recursion. Later work determined sharp bounds of order $X^{\log_4 3}$, periodic fluctuations, and extensions to generalized digit characters and other progressions [17, 10, 11]. The elementary dyadic formulas (143)–(144) exhibit the exponent and its two-scale oscillation before any deeper estimate is needed.

13 Combinatorics and dynamics of the infinite word

The arithmetic sign sequence and the binary word carry the same information, but the word language exposes repetition avoidance and dynamical properties that do not look like formulae for individual terms.

13.1 Uniform recurrence and aperiodicity

The substitution matrix of $\mu(0) = 01$, $\mu(1) = 10$ has every entry positive. Hence the substitution is primitive. Standard primitive-substitution theory implies that the fixed point is uniformly recurrent: every finite factor reappears with bounded gaps, and the associated shift orbit closure is minimal and uniquely ergodic [19]. In particular, every factor has a well-defined frequency; the two letters each have frequency $1/2$.

The word is not eventually periodic. Here is an elementary proof. Suppose an eventual period p existed. Removing powers of two from p using $\varepsilon_{2n} = \varepsilon_n$ would give an odd eventual period q . Applying that period at even indices gives, for all sufficiently large n ,

$$\varepsilon_n = \varepsilon_{2n} = \varepsilon_{2n+q} = -\varepsilon_{n+(q-1)/2}.$$

Applying this relation twice shows that $q - 1$ is also an eventual period. Since consecutive integers have greatest common divisor one, period one would follow, making the tail constant; this contradicts $\varepsilon_{2n+1} = -\varepsilon_n$.

13.2 Complement and reversal closure

Complementing letters commutes with the substitution up to exchanging its two seed letters. Therefore, whenever a finite word occurs, its bitwise complement also occurs. The dyadic reflection formula (15) similarly shows that the reversal of every prefix $\mu^m(0)$ is either that prefix or its complement. Passing to factors proves that the full factor language is closed under both complement and reversal.

The exact balance formula (110) is the one-letter shadow of this symmetry. For longer factors, unique ergodicity and complement invariance imply that complementary factors have equal frequencies.

13.3 Overlap avoidance and critical exponent

An *overlap* is a word of the form $axaxa$, where a is one letter and x is possibly empty. Thue proved that the fixed point of (10) is overlap-free [23]. Consequently it contains no cube www with $w \neq \emptyset$, and no factor of exponent strictly larger than two. On the other hand it contains squares of arbitrarily large length, obtained by iterating the uniform morphism on any square factor. Hence its critical exponent is exactly

$$\boxed{2}. \tag{148}$$

The overlap-free theorem is much stronger than the visible absence of 000 and 111. It was one of the original reasons for Thue’s study of the word and remains central in combinatorics on words.

13.4 Linear factor complexity

Let $p(n)$ be the number of distinct length- n factors. Uniform morphicity gives $p(n) = \mathcal{O}(n)$, while aperiodicity and the Morse–Hedlund theorem give $p(n) \geq n + 1$. Thus the word has linear, but nonminimal, factor complexity. Exact piecewise formulas and a complete description of square factors are known; see the survey literature in [2, 3]. This is another useful contrast: the word is far from periodic, yet its local language grows only linearly rather than exponentially.

14 Hankel determinants, Mahler values, and Diophantine consequences

14.1 Hankel determinants built from the signs

Define

$$\Delta_n = \det(\varepsilon_{i+j})_{0 \leq i, j < n}. \tag{149}$$

The first values are

$$1, -2, 4, 8, -16, -32, -64, 128, -256, -1536, \dots \quad (150)$$

Unlike the engineered Hessenberg determinant in [Theorem 7.5](#), these matrices use no ruler weights and have no evident triangular reduction.

Theorem 14.1 (Nonvanishing Hankel determinants; literature theorem). *For every $n \geq 1$,*

$$\boxed{\Delta_n \neq 0.} \quad (151)$$

This was proved by Allouche, Peyrière, Wen, and Wen through a finite system of determinant recurrences; later proofs include a short reduction modulo two and a combinatorial argument [\[1, 13, 7\]](#). Nonvanishing means that the coefficient series $E(z)$ is normal for diagonal Padé approximation at every order: no unexpected rank defect destroys the corresponding rational approximant. This determinant rigidity is one of the main bridges from automatic sequences to Diophantine approximation.

14.2 Transcendence of the function and of algebraic values

The natural boundary proved in [Section 7](#) immediately implies that $E(z)$ is transcendental over $\mathbb{C}(z)$: an algebraic function has only finitely many branch points and poles on the Riemann sphere and cannot have a natural boundary. Mahler’s method gives the much stronger arithmetic statement.

Theorem 14.2 (Mahler; literature theorem). *If α is algebraic and $0 < |\alpha| < 1$, then*

$$\boxed{E(\alpha) \text{ is transcendental.}} \quad (152)$$

Consequently $T(\alpha)$ is transcendental as well.

The hypotheses are particularly clean here: the forward orbit $\alpha, \alpha^2, \alpha^4, \dots$ remains in the open unit disk and never meets the singularity of the rational coefficient $1 - z$ in the Mahler equation [\(68\)](#). The original result is one of the foundational examples of Mahler’s method [\[15\]](#).

For every integer $b \geq 2$, put

$$\xi_b = T(1/b) = \sum_{n \geq 0} \frac{\tau(n)}{b^n} = b\kappa_b. \quad (153)$$

The theorem shows that ξ_b is transcendental. The quality of its rational approximations is known exactly.

Theorem 14.3 (Bugeaud; literature theorem). *For every integer $b \geq 2$, the irrationality exponent of ξ_b is*

$$\boxed{\mu(\xi_b) = 2.} \quad (154)$$

Here $\mu(\xi)$ is the supremum of the real numbers μ for which $|\xi - p/q| < q^{-\mu}$ has infinitely many rational solutions. The value two is the smallest possible for an irrational real number, by Dirichlet’s theorem. Thus the Thue–Morse–Mahler numbers are transcendental but not exceptionally well approximable. Bugeaud’s proof uses Padé approximants and the nonvanishing Hankel structure [\[6\]](#); later work placed this in a broader determinant framework [\[8\]](#).

15 The base- B root-of-unity family

The binary sequence is the first member of a general digit-character construction. Let $B \geq 2$, write $s_B(n)$ for the sum of the base- B digits of n , and fix $\zeta \in \mathbb{C}$. Define

$$u_n = \zeta^{s_B(n)}. \quad (155)$$

For $0 \leq n < B^m$, choosing the m digits independently gives the finite product

$$\sum_{n=0}^{B^m-1} u_n z^n = \prod_{j=0}^{m-1} \left(1 + \zeta z^{B^j} + \zeta^2 z^{2B^j} + \cdots + \zeta^{B-1} z^{(B-1)B^j} \right). \quad (156)$$

It also gives block multiplicativity

$$u_{B^m q + r} = u_q u_r \quad (0 \leq r < B^m). \quad (157)$$

Suppose now that ζ is a nontrivial B -th root of unity. At $z = e^t$, each factor in (156) has a simple zero at $t = 0$, because

$$\sum_{d=0}^{B-1} \zeta^d = 0, \quad \sum_{d=0}^{B-1} d \zeta^d = -\frac{B}{1-\zeta} \neq 0. \quad (158)$$

Thus the generalized character annihilates powers below degree m .

Theorem 15.1 (Base- B Prouhet character). *If $\zeta^B = 1$ and $\zeta \neq 1$, then*

$$\sum_{n=0}^{B^m-1} \zeta^{s_B(n)} n^r = 0 \quad (0 \leq r < m), \quad (159)$$

$$\sum_{n=0}^{B^m-1} \zeta^{s_B(n)} n^m = m! \left(-\frac{B}{1-\zeta} \right)^m B^{\binom{m}{2}}. \quad (160)$$

Proof. Set $z = e^t$ in (156). The derivative at zero of the j -th factor is $-B^{j+1}/(1-\zeta)$. Their product is the coefficient of t^m , namely $(-B/(1-\zeta))^m B^{\binom{m}{2}}$. Compare exponential generating coefficients. $\square$

For $B = 2$ and $\zeta = -1$, one recovers $u_n = \varepsilon_n$, the lacunary factors $1 - z^{2^j}$, and the exact moment (91). This generalization shows which parts of the atlas arise from digit independence and root-of-unity cancellation, and which depend specifically on binary parity, Lucas’s theorem modulo two, or two-adic valuations.

16 Formula selection and a Lean formalization roadmap

16.1 Which representation is useful where?

No single formula dominates all others. The right interface depends on the data already present in a proof or computation.

Representation	Required input	Best use
Binary popcount	machine word or bit list	Fast evaluation; executable normalization.
Even–odd recursion	quotient and low bit	Induction; automata; streaming generation.
Dyadic block law	high and low bit blocks	Kernel arguments; divide-and-conquer proofs.
XOR character	bitwise XOR	Boolean harmonic analysis; carry-free sums.
Factorial valuation	$n!$ or floor sums	Number-theoretic contexts already using valuations.

Representation	Required input	Best use
Central binomial valuation	$\binom{2n}{n}$	A one-integer formula; Kummer interfaces.
Carry cocycle	addition and a binomial valuation	Translating XOR identities to ordinary addition.
Pascal support	parity of row entries	Lucas-theorem and combinatorial arguments.
Modulo-three odd count	$\nu(n) \bmod 3$	Log-free version of the Pascal-row formula.
Finite lacunary product	polynomial or formal series	Prouhet, cyclotomic, Fourier, and coefficient proofs.
Ruler convolution	earlier signs and $v_2(k)$	Digit-free recurrence; determinant extraction.
Finite differences	arbitrary test function	Cancellation beyond monomials; positivity and bounds.
Walsh transform	Boolean vectors	One-frequency spectral normal form.
Ordinary DFT	roots of unity	Cyclic spectra and rarefied sums.
Riesz product	Fourier intensity	Spectral measures and autocorrelation.
Dirichlet/Mellin transform	analytic continuation	Global analytic fingerprints and special values.

16.2 Existing formalized spine

The repository’s proved development already contains the core chain

binary recursion $\implies$ block product $\implies$ Prouhet cancellation $\implies$ dyadic Fabius arithmetic. (161)

The principal files relevant to this article include

- `ThueMorseGenerating.lean`: finite block polynomial and lacunary product;
- `ThueMorseExponential.lean`: exponential form, order of vanishing, and moments;
- `ThueMorsePrefix.lean`: iterated-prefix identities;
- `ThueMorseApproximation.lean` and `PaperKFFoldThueMorse.lean`: convergence of discrete prefix constructions;
- `ThueMorseBinomialLog.lean`: the Pascal-row logarithm;
- `DyadicClosedForm.lean`, `DyadicCorrectness.lean`, and the `Fabius*QBinomial*.lean` files: exact dyadic evaluation and its q -binomial forms.

The main exposition also names `binaryWeight`, `thueMorseSign`, and the primitive-recursive bit evaluator `tmBitPR` as the foundational term-level objects.

16.3 Suggested dependency order for the new formulae

A practical formalization should avoid beginning with the analytic or determinant results. The following order maximizes reuse.

1. **Digit and bit API.** Prove floor extraction, XOR parity, block reflection, and the ruler transition from existing binary-weight lemmas.

2. **Valuations.** Establish the base-two Legendre identity and then central binomial, Catalan, Kummer, and multinomial corollaries.
3. **Pascal support.** Specialize Lucas modulo two, identify submasks, count odd entries, and prove the modulo-three and r -nomial formulas.
4. **Boolean incidence algebra.** Formalize the submask zeta transform and the explicit Thue–Morse Möbius inverse.
5. **Euler-transform layer.** Starting from the formal product, derive the ruler logarithm, convolution recurrence, Bell formula, and Hessenberg determinant.
6. **General finite differences.** Lift polynomial Prouhet cancellation to arbitrary commuting translations and derive sparse Pascal-row cancellation.
7. **Finite harmonic analysis.** Evaluate the block polynomial on roots of unity, prove the DFT support and Walsh delta theorem, then derive the modulo-three rarefied formulas.
8. **Analysis.** Add the Mellin continuation and parameterized infinite products only after the formal power-series and finite-sum layers are stable.

Natural module names would be

Proposed module	Principal declarations
<code>ThueMorseValuation.lean</code>	Legendre digit sum, central-binomial valuation, Catalan parity, carry cocycle.
<code>ThueMorsePascalSupport.lean</code>	Lucas submask criterion, Glaisher count, sparse columns, modulo-three formula.
<code>ThueMorseBooleanMobius.lean</code>	Submask incidence matrix and explicit inverse.
<code>ThueMorseEulerTransform.lean</code>	Ruler logarithm, convolution recurrence, Bell and determinant formulae.
<code>ThueMorseSparseProuhet.lean</code>	Arbitrary-support product and exact boundary moment.
<code>ThueMorseFiniteFourier.lean</code>	DFT product, even-frequency vanishing, Walsh transform.
<code>ThueMorseRarefied.lean</code>	Root-of-unity filter and exact modulo-three dyadic sums.
<code>ThueMorseDirichlet.lean</code>	Mellin representation, superpolynomial endpoint decay, entire continuation.

16.4 Executable checks

For evaluation, popcount parity remains the correct implementation. The other formulae are valuable as proof interfaces and independent regression oracles.

Listing 1: Cross-checking independent formula families

```

eps(n) :
  return +1 if popcount(n) is even else -1

for n from 0 to limit:
  e = eps(n)
  assert e == (-1)^(n - v2(factorial(n)))
  assert e == (-1)^v2(binomial(2*n, n))
  odd = count(k in 0..n where binomial(n,k) is odd)
  assert odd == 2^popcount(n)
  assert e == 3 - 2*(odd mod 3)

for m from 0 to maxLevel:
  P = sum(n in 0..2^m-1, eps(n)*z^n)
  assert P == product(j in 0..m-1, 1-z^(2^j))

```

```

for r from 0 to m-1:
    assert sum(n in 0..2^m-1, eps(n)*n^r) == 0
assert sum(n in 0..2^m-1, eps(n)*n^m)
    == (-1)^m * factorial(m) * 2^(m*(m-1)/2)

```

For fixed-width machine integers, the optimal evaluator is a parity-popcount instruction or a sequence of XOR folds. The recursion is optimal for an unbounded inductive natural. The central-binomial and Pascal formulas should not be used to compute large terms literally; their role is to expose mathematical structure and to connect with existing arithmetic data.

17 Worked examples and consistency checks

17.1 One index through ten formulae

Take $n = 13 = (1101)_2$. Its weight is three, so $\tau(13) = 1$ and $\varepsilon_{13} = -1$. The same value emerges as follows.

1. **Digit product:** $(1-2)(1-0)(1-2)(1-2) = -1$.
2. **Floor sum:** $\lfloor 13/2 \rfloor + \lfloor 13/4 \rfloor + \lfloor 13/8 \rfloor = 6 + 3 + 1 = 10$, so $(-1)^{13-10} = -1$.
3. **Factorial valuation:** $v_2(13!) = 10$, again giving $(-1)^3$.
4. **Central binomial:** $v_2\binom{26}{13} = s_2(13) = 3$.
5. **Catalan:** $v_2(C_{13}) = s_2(14) - 1 = 2$ and $v_2(14) = 1$, whose sum is three.
6. **Sparse Pascal columns:** $\binom{13}{1}, \binom{13}{4}, \binom{13}{8}$ are odd, while $\binom{13}{2}$ is even; the parity sum is three.
7. **Pascal support:** row 13 has $2^3 = 8$ odd entries, and $8 \bmod 3 = 2$, so $3 - 2(2) = -1$.
8. **Submask product:**

$$\sum_{k \preceq_2 13} \varepsilon_k z^k = (1-z)(1-z^4)(1-z^8),$$

whose coefficient at z^{13} is -1 .

9. **Ruler transition:** starting at $\varepsilon_0 = 1$, repeatedly apply $\varepsilon_{j+1} = -(-1)^{v_2(j+1)}\varepsilon_j$ through $j = 12$.
10. **Hessenberg determinant:** with $a_j = 2^{v_2(j)+1} - 1$, $\det H_{13} = (-1)^{13} 13! \varepsilon_{13} = 13!$.

The routes use radically different intermediate objects, but every one reduces to the same binary parity character.

17.2 A complete four-bit block

At $m = 4$,

$$\begin{aligned}
 P_4(z) &= 1 - z - z^2 + z^3 - z^4 + z^5 + z^6 - z^7 \\
 &\quad - z^8 + z^9 + z^{10} - z^{11} + z^{12} - z^{13} - z^{14} + z^{15} \\
 &= (1-z)(1-z^2)(1-z^4)(1-z^8).
 \end{aligned} \tag{162}$$

The factor $(1-z)^4$ divides this polynomial, so its first four ordinary derivatives at one vanish through order three. Equivalently,

$$\sum_{n < 16} \varepsilon_n n^r = 0 \quad (0 \leq r < 4), \quad \sum_{n < 16} \varepsilon_n n^4 = 4! 2^6 = 1536. \tag{163}$$

The sign is positive because $(-1)^4 = 1$.

The residue-class sums modulo three are, from (143),

$$\left(\sum_{\substack{n < 16 \\ n \equiv 0(3)}} \varepsilon_n, \sum_{\substack{n < 16 \\ n \equiv 1(3)}} \varepsilon_n, \sum_{\substack{n < 16 \\ n \equiv 2(3)}} \varepsilon_n \right) = (6, -3, -3). \quad (164)$$

Their sum is zero, as required by the ordinary block balance, but the distribution among residue classes is highly nonuniform.

17.3 A carry-corrected addition

Take $a = 13 = (1101)_2$ and $b = 7 = (0111)_2$. Then $\varepsilon_{13} = \varepsilon_7 = -1$, while $a + b = 20 = (10100)_2$ has sign $+1$. Kummer’s formula gives

$$v_2 \binom{20}{13} = s_2(13) + s_2(7) - s_2(20) = 3 + 3 - 2 = 4. \quad (165)$$

The correction factor $(-1)^4 = 1$ precisely restores multiplicativity. This example also illustrates why a yes/no statement about the presence of carries is insufficient: their parity, with propagation counted, is the invariant required by the sign.

18 Concluding synthesis

The Thue–Morse sequence is often introduced by the sentence “take the parity of the number of one-bits.” Correct as that is, it conceals the main structural fact. The same parity character is simultaneously

- a two-state automatic sequence and a character of the Boolean XOR group;
- the parity of a factorial, central-binomial, Catalan, or carry valuation;
- the Möbius kernel of the Boolean support of Pascal’s triangle;
- the coefficient sequence of a lacunary Mahler product;
- an inclusion–exclusion kernel for the dyadic differences $1, 2, 4, \dots$;
- Prouhet’s exact annihilator of low-degree polynomials;
- one Walsh frequency but a singular continuous ordinary Fourier object;
- the sign system in the global Fabius translate expansion and in exact dyadic cancellation;
- the source of rarefied arithmetic-progression biases, Hankel rigidity, and transcendental Mahler values.

These are not unrelated coincidences. Binary digit independence gives the finite product; Lucas turns that independence into Pascal support; Legendre and Kummer turn it into valuations and carries; evaluating the product at 1, roots of unity, exponentials, or translation operators yields Prouhet, Fourier, rarefied, and finite-difference identities. The binomial–logarithm formula sits exactly where the Pascal and digit branches meet.

The deepest reusable principle is therefore not any one boxed expression, but the following commuting diagram of transformations:

$$\begin{array}{c}
 \text{binary digits} \longleftrightarrow \text{Boolean character} \longleftrightarrow \text{Lucas support,} \\
 \text{binary digits} \longleftrightarrow \text{Legendre–Kummer valuations,} \\
 \text{Boolean character} \longleftrightarrow \prod_j (1 - z^{2^j}) \longrightarrow \left\{ \begin{array}{l} \text{finite differences and Prouhet moments,} \\ \text{ordinary Fourier and Riesz products,} \\ \text{Mahler, Mellin, and Dirichlet transforms,} \\ \text{root-of-unity filters and rarefied sums.} \end{array} \right. \quad (166)
 \end{array}$$

For the Fabius project, this diagram provides several independent proof interfaces to the same arithmetic layer. For future formalization, it also suggests a clean order: digits, valuations and Lucas support first; formal products and finite differences next; spectral and analytic consequences last.

A The first 64 terms

The following table gives $\tau(n)$. Replacing each zero by $+1$ and each one by -1 gives ε_n .

start	+0	+1	+2	+3	+4	+5	+6	+7
0	0	1	1	0	1	0	0	1
8	1	0	0	1	0	1	1	0
16	1	0	0	1	0	1	1	0
24	0	1	1	0	1	0	0	1
32	1	0	0	1	0	1	1	0
40	0	1	1	0	1	0	0	1
48	0	1	1	0	1	0	0	1
56	1	0	0	1	0	1	1	0

The first 32-bit block is followed by its complement. Inside each 16-bit and 8-bit block, the same copy/complement rule recurses. This nested visual structure is precisely (11).

B A compact identity map

For quick reference, the main term-level identities are

$$\begin{aligned}
 \varepsilon_n &= (-1)^{s_2(n)} = \prod_{j \geq 0} (1 - 2b_j(n)) \\
 &= (-1)^{n - \sum_{j \geq 1} \lfloor n/2^j \rfloor} = (-1)^{n - v_2(n!)} \\
 &= (-1)^{v_2\binom{2n}{n}} = (-1)^{\sum_{j \geq 0} \binom{n}{2^j}} \\
 &= \prod_{j \geq 0} \cos\left(\pi \binom{n}{2^j}\right) = \prod_{j \geq 0} \cos\left(\pi \left\lfloor \frac{n}{2^j} \right\rfloor\right) \\
 &= \prod_{j \geq 0} \operatorname{sgn} \sin\left(\frac{\pi(n + 1/2)}{2^j}\right) \\
 &= (-1)^{\log_2 \nu(n)} = 3 - 2[\nu(n)]_3 \\
 &= [z^n] \prod_{j \geq 0} (1 - z^{2^j}) = \frac{(-1)^n}{n!} \det H_n.
 \end{aligned}$$

The principal block identities are

$$\begin{aligned}
\sum_{n < 2^m} \varepsilon_n z^n &= \prod_{j < m} (1 - z^{2^j}), \\
\sum_{n < 2^m} \varepsilon_n e^{nt} &= \prod_{j < m} (1 - e^{2^j t}), \\
\sum_{n < 2^m} \varepsilon_n f(x + nh) &= (-1)^m \Delta_h \Delta_{2h} \cdots \Delta_{2^{m-1}h} f(x), \\
\sum_{n < 2^m} \varepsilon_n n^r &= 0 \quad (r < m), \\
\sum_{n < 2^m} \varepsilon_n n^m &= (-1)^m m! 2^{\binom{m}{2}}, \\
\widehat{\varepsilon}_m(k) &= \prod_{j < m} (1 - e^{-2\pi i k 2^j / 2^m}), \\
2^{-m} |P_m(e^{2\pi i x})|^2 &= \prod_{j < m} (1 - \cos(2\pi 2^j x)).
\end{aligned}$$

Each line is obtained from the same finite polynomial by coefficient extraction, substitution, differentiation, or a change of harmonic-analysis basis.

C Consolidation provenance

The four source drafts had substantial duplicated text but also distinct mathematical contributions. The consolidation used the following division of labor.

- `Further_Formulae_for_the_Thue_Morse_Sequence-1.tex` supplied the strongest Boolean Möbius-inversion, ruler-convolution, Bell-polynomial, Hessenberg-determinant, finite-Fourier, and Dirichlet-transform material.
- `...-2.tex` supplied the master-catalogue organization, the carry-corrected addition law, mixed finite differences, all-order moment formulas, computational comparison, and the base- B discovery pattern.
- `...-3.tex` supplied the r -nomial odd-coefficient family, sparse Prouhet cancellation on arbitrary Pascal support, the characteristic-two algebraic series, and a particularly clean separation between elementary proofs and literature results.
- `...-4.tex` supplied the logarithm-free modulo-three Pascal formula, odious and evil enumerators, the integer algebraic lift, autocorrelation and Riesz-product organization, and several exact integral and product forms.

The present article adopts one notation, one theorem numbering system, one bibliography, and one proof-status convention. Repeated elementary arguments were retained only once, while stronger versions replaced weaker duplicates. Sections on rarefied sums, combinatorics on words, Hankel determinants, natural boundaries, Mahler values, and irrationality exponents were added during consolidation.

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